

WILLOW GROVE NAS, PA, USA

WMO: 724086

Lat: 40.200N

Lon: 75.150W

Elev: 361

StdP: 14.51

Time Zone: -5.00 (NAE)

Period: 90-11

WBAN: 14793

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	16.6	-4.7	4.3	16.5	-0.8	5.3	20.3	23.7	31.4	20.4	32.9	7.4	300	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	91.9	73.9	89.4	72.6	86.8	71.5	77.1	87.5	75.5	84.9	74.1	82.9	6.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	127.9	82.5	72.5	121.9	80.8	71.3	117.0	79.4	40.7	87.3	39.1	85.0	37.7	82.8	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.4	16.9	14.7	4.4	96.2	4.1	2.6	1.5	98.1	-0.9	99.6	-3.2	101.1	-6.1	103.0	
			DB	4.4	96.2	4.1	2.6	1.5	98.1	-0.9	99.6	-3.2	101.1	-6.1	103.0
			WB	3.1	80.6	3.7	2.5	0.5	82.4	-1.7	83.8	-3.8	85.2	-6.4	87.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	33.0	34.7	42.4	52.9	62.1	71.7	75.9	73.8	66.1	55.6	45.1	37.1
	DBStd	17.00	9.76	9.47	9.32	8.98	7.54	6.50	5.33	4.98	7.04	8.15	8.76	8.08
	HDD50	1948	534	433	273	66	2	0	0	0	0	32	201	407
	HDD65	4935	992	848	704	378	149	17	1	3	71	310	598	864
	CDD50	3525	7	6	37	154	379	651	802	737	484	207	53	8
	CDD65	1034	0	0	2	16	61	218	338	275	104	20	0	0
	CDH74	9298	0	0	35	192	659	2024	3167	2336	767	113	5	0
	CDH80	3237	0	0	6	64	205	718	1253	788	191	12	0	0
Wind	WSAvg	5.9	6.8	7.7	7.6	7.3	5.9	5.1	4.7	4.2	4.6	4.9	5.9	6.6
Precipitation	PrecAvg	47.7	3.6	2.8	4.0	3.8	4.8	4.1	4.3	4.1	3.9	3.2	3.9	3.8
	PrecMax	69.0	13.7	6.8	8.4	9.7	9.5	14.0	11.9	9.7	9.2	7.0	10.8	8.4
	PrecMin	30.6	0.7	0.8	1.6	0.7	1.3	0.5	1.0	1.1	1.4	1.2	0.5	0.7
	PrecStd	9.1	2.5	1.3	1.7	1.8	2.4	3.0	2.6	2.1	2.2	1.3	2.6	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	65.7	79.3	87.6	90.3	93.9	96.9	93.0	89.1	81.5	73.3	63.9
		MCWB	60.0	53.1	64.7	63.8	71.1	75.0	76.3	73.8	71.5	68.9	59.9	59.2
	2%	DB	59.4	59.7	69.5	79.6	85.4	90.4	92.9	89.9	84.9	77.4	68.4	59.1
		MCWB	54.8	51.2	57.2	61.3	67.5	73.3	74.5	72.9	70.4	66.2	59.0	54.8
	5%	DB	52.3	54.1	62.6	73.6	81.3	87.1	89.7	87.2	81.7	73.3	64.0	53.8
		MCWB	47.5	45.6	52.5	58.3	66.7	71.4	73.6	72.3	70.0	63.9	57.6	48.6
	10%	DB	46.6	49.1	57.0	68.5	77.0	84.2	86.7	84.6	78.7	69.8	59.9	49.7
		MCWB	41.7	42.5	48.9	55.9	62.6	69.4	72.0	70.7	67.9	61.7	53.3	44.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.6	58.4	66.1	67.4	75.5	78.2	79.9	78.2	76.0	71.3	65.0	60.5
		MCDB	64.0	62.8	77.5	79.5	86.6	88.9	92.0	87.4	84.6	76.8	69.7	63.2
	2%	WB	55.5	51.5	59.6	63.6	70.9	75.6	77.7	76.0	74.0	68.7	61.8	55.5
		MCDB	59.4	58.4	66.3	76.1	81.7	86.3	88.6	85.3	81.0	75.2	65.4	58.7
	5%	WB	48.1	47.2	54.3	60.7	68.0	73.8	76.0	74.5	72.0	66.4	58.0	49.8
		MCDB	51.6	53.3	60.5	71.2	78.2	83.7	85.5	83.2	78.6	71.4	64.1	53.1
	10%	WB	42.2	42.7	49.9	57.5	65.1	71.9	74.3	72.9	69.7	63.1	53.6	45.5
		MCDB	45.9	49.0	56.4	66.6	74.0	80.7	83.1	81.1	76.0	68.8	59.4	48.4
Mean Daily Temperature Range	5% DB	MDBR	14.0	15.8	17.5	19.7	20.4	19.5	18.3	18.8	18.6	19.7	16.6	14.2
		MCDBR	20.3	22.6	25.6	25.9	26.2	22.8	21.5	21.1	20.6	22.0	21.8	20.3
		MCWBR	18.0	17.4	17.2	14.0	14.1	11.3	9.6	10.0	10.8	13.5	17.1	17.5
	5% WB	MCDBR	19.3	20.8	23.2	22.6	23.2	20.6	19.2	18.8	18.0	18.6	19.4	18.5
		MCWBR	18.6	17.8	17.8	15.1	14.2	11.4	10.2	10.7	11.4	13.2	17.7	18.1
Clear-Sky Solar Irradiance	taub	0.307	0.319	0.357	0.388	0.434	0.493	0.492	0.486	0.401	0.356	0.324	0.306	
	taud	2.460	2.413	2.375	2.317	2.215	2.090	2.108	2.111	2.321	2.457	2.512	2.514	
	Ebn at Noon	272	285	284	281	269	252	251	249	266	268	264	264	
	Edh at Noon	25	30	35	39	45	51	49	48	36	29	24	22	
All-Sky Solar Radiation	RadAvg	599	887	1171	1519	1691	1850	1892	1638	1361	964	682	516	
	RadStd	52	97	93	138	198	153	91	105	155	101	61	54	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+122

Nomenclature: See separate page